The Three Dimensions of Cognition
Intelligence, Knowledge, and Closure Capacity

Joachim De Witte

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Abstract

Contemporary discussions of intelligence, artificial intelligence (AI), artificial general intelligence (AGI), and artificial superintelligence (ASI) often rely on implicit one-dimensional views of cognition in which intelligence is treated as the primary scaling variable. Such perspectives struggle to distinguish between fundamentally different sources of cognitive capability, including reasoning ability, accumulated knowledge, memory resources, parallelism, and architectural organization.

This paper proposes that cognition is more naturally described by three largely orthogonal dimensions: *intelligence*, representing the capacity for reasoning, abstraction, and pattern recognition; *knowledge*, representing the information available within a cognitive system; and *closure capacity*, representing the structural capabilities of the underlying cognitive closure, including memory scale, communication bandwidth, parallelism, coordination mechanisms, and the ability to extend, merge, or reorganize cognitive processes.

Building on this framework, we introduce a distinction between *Biological Cognition* (BC), emerging from relatively fixed biological closures, and *Architectural Cognition* (AC), emerging from engineered closures whose structure can be expanded, distributed, or reconfigured. We further examine *Hybrid Cognition* (HC) as a form of cognition that combines multiple closures and thereby partially transcends the limitations of individual cognitive systems.

The framework provides an alternative interpretation of cognitive scaling and offers a possible explanation for the persistent ambiguity surrounding AI, AGI, and ASI terminology. Rather than viewing cognitive progress solely as movement along an intelligence axis, the proposed model argues that advances frequently occur through increases in knowledge availability and closure capacity. Finally, the paper discusses how closure dynamics may influence the emergence of subjective time perception through differences in internal cognitive evolution, providing a conceptual bridge between cognitive architecture and temporal experience.

1 Introduction

Discussions of cognition, intelligence, artificial intelligence (AI), artificial general intelligence (AGI), and artificial superintelligence (ASI) frequently rely on an implicit assumption: cognitive capability can be represented along a single dimension, usually described as intelligence. Systems are then compared according to their perceived position on this scale, leading to classifications such as “less intelligent”, “human-level intelligent”, or “superintelligent”.

While this perspective is intuitive, it struggles to explain many commonly observed phenomena. A modern scientist and a medieval scholar may possess similar biological cognitive substrates while exhibiting vastly different capabilities due to differences in available knowledge. Likewise, a team of scientists often outperforms a single expert despite no corresponding increase in individual intelligence. Contemporary AI systems further complicate the picture. Large language models, agentic workflows,

retrieval systems, distributed memory architectures, and multi-agent environments often achieve substantial performance gains through increased memory, communication, parallel processing, and access to information rather than through clearly measurable increases in reasoning ability alone.

These observations suggest that cognition may not be adequately characterized by intelligence as a single variable. Instead, multiple independent dimensions may contribute to cognitive capability. This paper proposes a simple conceptual framework in which cognition is described by three largely orthogonal dimensions:

1. **Intelligence**, representing the ability to reason, learn, abstract, generalize, and recognize patterns.
2. **Knowledge**, representing the information, experience, memories, and representations available within a cognitive system.
3. **Closure Capacity**, representing the structural capabilities of the underlying cognitive closure, including memory scale, communication bandwidth, parallelism, coordination mechanisms, and the ability to extend, merge, distribute, or reorganize cognitive processes.

The central thesis of this paper is that closure capacity constitutes an independent cognitive dimension that is often underrepresented in discussions of both biological and artificial cognition. Many debates surrounding AI, AGI, and ASI implicitly combine intelligence, knowledge, and closure capacity into a single measure, making precise definitions difficult and frequently leading to conceptual ambiguity.

Building upon this framework, we distinguish between *Biological Cognition* (BC), which emerges from relatively fixed biological closures, and *Architectural Cognition* (AC), which emerges from engineered closures whose structure may be expanded, distributed, replicated, merged, or reconfigured. We further introduce *Hybrid Cognition* (HC) as a form of cognition arising from the cooperation of multiple closures, allowing cognitive systems to partially transcend the limitations of individual substrates.

Finally, the paper explores several implications of this framework. These include alternative perspectives on cognitive scaling, a possible explanation for the persistent ambiguity of AGI and ASI definitions, and the hypothesis that differences in closure dynamics may influence subjective time perception through variations in internal cognitive evolution. While the framework is conceptual in nature, it aims to provide a unified vocabulary for discussing cognition across biological, architectural, and hybrid systems.

2 Related Work Within Ecosystem

The framework proposed in this paper emerges from a broader sequence of investigations within the Triadic Cosmos ecosystem concerning intelligence, cognition, closure dynamics, and temporal structure.

"The Triadic Cosmos: Emergence of Intelligence" introduced a substrate-independent model of intelligence based on four structural dimensions: multiplicity, informational richness, recursion depth, and energetic capacity [1]. That work examined the conditions under which intelligent behavior emerges across biological, artificial, and hybrid substrates, providing a structural account of intelligence that is independent of specific implementations.

"A Structural Taxonomy of Intelligence" extended this analysis by proposing a classification framework for different intelligent systems [2]. Using the IEGR structural model, the taxonomy distinguished between various cognitive regimes and argued that observed differences between intelligent systems often arise from structural completeness and scaling characteristics rather than intelligence alone.

"Architectural Intelligence Theory" introduced the concepts of Architectural General Intelligence and Architectural Super Intelligence [3]. That work emphasized the role of cognitive closure, modularity, and architectural organization, arguing that cognition cannot be adequately understood through statistical capability alone. The distinction between biological, architectural, and closure-breaking cognitive systems provides an important conceptual foundation for the present work.

"The Lifecycle Architecture of Synthetic Cognition" shifted attention from static cognitive capability toward cognitive dynamics [4]. Through the study of attractor formation, senescence, collapse, and lifecycle evolution in synthetic cognitive systems, it was argued that cognition is fundamentally temporal

and closure-bound. The work introduced the idea that cognitive systems evolve within finite closures whose properties change over time, providing a developmental view of cognition beyond static intelligence measures.

Finally, "Dual Time in Nature" introduced the distinction between shared external time and closure-bound material time [5]. That framework proposed that different closures may generate and experience internal temporal evolution at different rates, linking closure structure to subjective temporal experience across biological, physical, and artificial systems.

The present paper integrates these lines of work into a unified cognitive framework. Whereas earlier ecosystem contributions focused primarily on the emergence of intelligence, the taxonomy of intelligent systems, architectural organization, lifecycle dynamics, or closure-dependent temporality, the current work focuses explicitly on cognition itself.

Three dimensions are proposed: intelligence, knowledge, and closure capacity. Intelligence represents the system's reasoning capability; knowledge represents the information available within the closure; and closure capacity represents the structural properties that determine the scale, depth, and evolutionary potential of cognition.

Within this interpretation, the previous ecosystem works can be viewed as investigating different aspects of the same underlying phenomenon. The Emergence of Intelligence paper explored the formation of intelligence; the Taxonomy paper explored the classification of intelligent systems; the Architectural Intelligence paper explored closure structure; the Lifecycle paper explored closure evolution; and the Dual Time paper explored closure-bound temporality. The present work unifies these perspectives by positioning intelligence, knowledge, and closure capacity as the three primary dimensions through which cognition may be analyzed across biological, architectural, and hybrid systems.

3 Related Work

The present framework intersects with several established research traditions within cognitive science, artificial intelligence, systems theory, and the philosophy of mind. Although these traditions differ substantially in their assumptions and methodologies, many have recognized that cognition cannot be reduced to a single measure of intelligence.

3.1 Multidimensional Views of Cognition

Traditional intelligence research has often focused on the measurement of general cognitive ability through concepts such as the general intelligence factor (g). At the same time, alternative frameworks have argued that cognition is inherently multidimensional. Examples include Gardner's theory of multiple intelligences, Sternberg's triarchic theory of intelligence, and later cognitive frameworks that distinguish between memory, reasoning, attention, learning, and executive function.

The present work shares the broader intuition that cognition cannot be captured by a single scalar quantity. However, rather than decomposing cognition into psychological abilities, the proposed framework focuses on three structural dimensions: intelligence, knowledge, and closure capacity.

3.2 Cognitive Architectures

Research on cognitive architectures such as SOAR, ACT-R, Sigma, and related frameworks has long emphasized the importance of memory systems, procedural knowledge, representational structures, and control mechanisms in cognitive behavior.

These architectures demonstrate that comparable reasoning capabilities may produce very different cognitive performance depending on memory resources, architectural design, and information availability. The present framework is compatible with these observations and generalizes them through the concept of closure capacity, which captures the structural properties that influence the scope and evolution of cognition.

3.3 Extended and Distributed Cognition

Theories of extended cognition and distributed cognition argue that cognitive processes are not necessarily confined to a biological brain. The extended mind framework of Clark and Chalmers and later work on distributed cognition, collective intelligence, and human-computer interaction suggest that tools, environments, and collaborative systems can become integral components of cognition.

These perspectives strongly resonate with the present framework. Hybrid cognition is interpreted here as a form of cognition emerging across multiple interacting closures, allowing cognitive capability to expand through shared knowledge, external memory, cooperation, and increased closure capacity.

3.4 Collective and Ecosystem Intelligence

Research on collective intelligence, swarm intelligence, organizational cognition, and multi-agent systems has demonstrated that groups can exhibit capabilities that exceed those of individual participants. Studies of human teams, distributed organizations, and cooperative AI systems increasingly suggest that intelligence may emerge at population and ecosystem scales.

The present work extends this perspective by proposing that cognition itself can be analyzed at multiple closure levels, ranging from individual biological organisms to architectural and hybrid ecosystems. This view aligns with broader trends toward understanding intelligence as an emergent property of interacting systems rather than isolated agents.

3.5 Artificial General Intelligence and Cognitive Scaling

Contemporary AGI discussions often focus on scaling laws, model size, computational resources, and benchmark performance. While these approaches provide important insights into capability growth, they frequently treat intelligence as the primary explanatory dimension.

The framework proposed in this paper argues that capability differences may also arise from variations in knowledge availability and closure capacity. This perspective suggests that apparent progress toward AGI or ASI may involve multiple forms of scaling simultaneously, including intelligence scaling, knowledge scaling, and closure-capacity scaling.

3.6 Time Perception and Cognitive Dynamics

Research in psychology and neuroscience has investigated subjective time perception in relation to attention, novelty, memory formation, emotional state, and information processing. Several theories propose that experienced time is linked to the amount of cognitive activity or change occurring within a system.

The present framework is broadly compatible with these observations but approaches the problem from a structural perspective. Building upon closure-based models of cognition, it explores the possibility that subjective time emerges from the accumulation of internal cognitive evolution occurring within a closure and that differences in closure capacity may influence the generation of material time.

3.7 Position of the Present Work

Unlike most existing approaches, the framework proposed here does not seek to define cognition exclusively through intelligence, memory, consciousness, symbol manipulation, embodiment, or computational scale. Instead, it proposes that cognition can be understood through three interacting dimensions: intelligence, knowledge, and closure capacity.

The resulting perspective is intended as a substrate-independent conceptual framework applicable to biological cognition, architectural cognition, and hybrid cognition alike.

4 Current limitations of intelligence-centric cognition models

Many contemporary discussions of cognition implicitly assume that cognitive capability can be approximated by a single dominant variable: intelligence. Within both cognitive science and artificial intelligence research, systems are frequently compared according to perceived differences in reasoning ability, pattern recognition, abstraction, problem-solving performance, or learning capacity. While such measures capture important aspects of cognition, they provide only a partial description of why cognitive systems differ in practice.

Intelligence-centric models encounter difficulties when explaining situations in which cognitive performance changes substantially despite no obvious increase in reasoning ability. Consider a modern scientist and a scholar from the medieval period. While both individuals possess broadly similar biological cognitive substrates, the modern scientist benefits from centuries of accumulated scientific knowledge, formal education, mathematical frameworks, technological tools, and access to global information networks. In many practical situations, the modern scientist may appear vastly more capable, yet this difference cannot be attributed solely to intelligence.

A similar observation arises in collective cognition. Research teams, organizations, and scientific communities routinely solve problems that exceed the capacity of any individual member. The resulting increase in cognitive capability emerges not from increased individual intelligence but from enhanced coordination, communication, knowledge sharing, and specialization. In such cases, cognitive performance appears to scale through structural properties of the system rather than through changes in intelligence alone.

Recent developments in artificial intelligence further expose this limitation. Large language models often achieve significant performance improvements through access to larger training corpora, extended memory systems, retrieval mechanisms, external tools, agentic workflows, and multi-agent architectures. While some improvements may reflect increases in reasoning capability, many arise through expanded access to information and through architectural mechanisms that increase the effective capacity of the cognitive system.

These observations suggest that treating cognition as a one-dimensional intelligence scale may obscure important distinctions. In practice, discussions of AI, AGI, and ASI frequently combine multiple factors under the single label of intelligence. Knowledge accumulation, memory size, communication bandwidth, parallel processing, external tooling, and collaborative structures are often viewed as manifestations of intelligence despite representing qualitatively different properties.

This conceptual conflation contributes to persistent ambiguity in contemporary AI discourse. For example, a system may appear more capable because it possesses greater knowledge, because it can process larger contexts, because it can coordinate multiple specialized subcomponents, or because it exhibits stronger reasoning abilities. When these factors are not clearly separated, terms such as "human-level intelligence", "general intelligence", and "superintelligence" become difficult to define rigorously.

The central limitation of intelligence-centric cognition models is therefore not that intelligence lacks importance, but that intelligence alone does not fully characterize cognition. A broader framework is required to distinguish between the ability to reason, the knowledge available within a cognitive system, and the structural properties of the closure that supports cognitive activity. The following section introduces these dimensions explicitly and proposes a three-dimensional model of cognition based on intelligence, knowledge, and closure capacity.

5 Cognition as three orthogonal dimensions

The previous section argued that intelligence alone does not adequately explain differences in observed cognitive capability. Similar reasoning abilities may produce radically different outcomes depending on the knowledge available to a cognitive system and the structural properties of the closure within which that cognition operates. To address this limitation, we propose a three-dimensional model of cognition consisting of intelligence, knowledge, and closure capacity.

The model does not claim that these dimensions are perfectly independent. Interactions between them clearly exist. Greater closure capacity may facilitate the acquisition of knowledge, while intelligence influences how effectively knowledge can be acquired and utilized. Nevertheless, the dimensions are sufficiently distinct to justify separate consideration.

5.1 Intelligence in Closure

Intelligence describes the ability of a cognitive system to recognize patterns, form abstractions, reason about internal and external states, generate predictions, and solve problems. It represents the cognitive processes that transform information into understanding and action.

Examples of intelligence include:

- Logical reasoning.
- Pattern recognition.
- Learning and adaptation.
- Generalization from limited information.
- Planning and decision making.

Within the present framework, intelligence operates within a closure. It does not describe what information is available to the system, but rather how effectively the system can process available information.

Two cognitive systems may therefore possess similar knowledge while exhibiting different levels of intelligence due to differences in reasoning capability.

5.2 Knowledge in Closure

Knowledge describes the information, representations, memories, experiences, and learned structures available within a cognitive system. Knowledge does not directly represent processing ability, but rather the content upon which cognitive processes operate.

Examples include:

- Formal education.
- Long-term memory.
- Scientific knowledge.
- Learned representations.
- Training data.
- Cultural and historical information.

A modern scientist possesses access to vastly more accumulated knowledge than a medieval scholar, despite both operating within broadly similar biological cognitive substrates. Such differences illustrate how cognitive capability can increase through knowledge accumulation even in the absence of significant changes in the underlying intelligence of the system.

Intelligence determines how knowledge can be utilized, but knowledge itself represents an independent source of cognitive capability.

5.3 Closure Capacity

Closure capacity describes the structural capabilities of the closure that supports cognition. While intelligence concerns processing and knowledge concerns content, closure capacity concerns the capabilities and constraints of the cognitive substrate itself.

Closure capacity includes factors such as:

- Memory capacity.
- Context capacity.
- Communication bandwidth.
- Parallel processing capability.
- Coordination mechanisms.
- Access to external tools.
- The ability to replicate, merge, or reorganize cognitive processes.

This dimension is particularly important because it captures capabilities that cannot be adequately described as either intelligence or knowledge.

For example, a human researcher and a distributed multi-agent system may exhibit similar reasoning capabilities and possess comparable domain knowledge, yet the multi-agent system may solve larger problems by distributing work across multiple concurrent processes. The difference arises primarily from closure capacity rather than intelligence.

Similarly, access to external memory systems, computational tools, retrieval mechanisms, or collaborative cognitive structures often increases cognitive performance even when neither intelligence nor knowledge is fundamentally altered.

5.4 A Three-Dimensional Cognitive Space

Together, the three dimensions define a conceptual cognitive space:

$$\text{Cognition} = f(\text{Intelligence}, \text{Knowledge}, \text{Closure Capacity})$$

Within this space, cognitive systems may achieve similar observed performance through different combinations of dimensions. One system may compensate for limited intelligence through extensive knowledge, while another may compensate for limited knowledge through greater reasoning ability. Likewise, increased closure capacity may enable levels of performance unattainable by either intelligence or knowledge alone.

The framework therefore rejects the notion that cognition can be projected onto a single intelligence axis. Instead, cognitive capability emerges from the interaction of multiple dimensions that contribute independently to the overall behavior of a cognitive system.

The following sections apply this framework to biological, architectural, and hybrid forms of cognition, demonstrating how different closure structures produce distinct cognitive characteristics despite potentially similar levels of intelligence or knowledge.

6 Biological Closures and Biological Cognition

Within the proposed framework, a *Biological Closure* is a cognitive closure whose substrate is a living biological system. Human cognition provides the most familiar example, although the concept extends naturally to animals and biological organisms exhibiting information processing, adaptation, learning, and memory.

A biological closure establishes a persistent boundary between internal and external states. Sensory information enters the closure through biological interfaces, is processed by internal cognitive mechanisms, and contributes to the formation of memories, representations, and behavioral responses. Although biological cognition continuously exchanges information with its environment, its internal state remains largely inaccessible to external observers except through indirect communication.

A defining characteristic of biological closures is the relatively fixed nature of the underlying cognitive substrate. Biological cognition may evolve through learning, development, and experience, yet the physical structure supporting that cognition remains constrained by the properties of the biological organism. Biological closures therefore exhibit strong continuity across time, forming coherent and persistent identities.

Several properties commonly associated with biological cognition emerge naturally from these closure constraints:

- A single persistent identity.
- Continuous memory accumulation throughout life.
- Incremental learning and adaptation.
- Strong embodiment within a physical environment.
- Limited memory capacity.
- Limited communication bandwidth.
- Inability to directly replicate, merge, or parallelize cognitive processes.

These characteristics are often treated as intrinsic properties of cognition itself. Within the present framework, however, they are interpreted as consequences of biological closure structure rather than universal properties of all possible cognitive systems.

The fixed nature of biological closures produces both advantages and limitations. Persistent identity supports long-term coherence, accountability, and stable accumulation of experience. Continuous embodiment enables rich interaction with physical environments and provides grounding for perception and action. At the same time, biological cognition remains subject to constraints imposed by metabolism, aging, memory limitations, and finite processing resources.

Importantly, biological cognition demonstrates substantial variation in both intelligence and knowledge despite operating within broadly similar closure structures. Human populations exhibit wide ranges of reasoning capability, expertise, creativity, and accumulated knowledge while sharing many common biological constraints. This observation further supports the distinction between intelligence, knowledge, and closure capacity as separate dimensions of cognition.

Within the three-dimensional model introduced in the previous section, biological cognition may therefore be represented as a family of cognitive systems sharing similar closure characteristics while differing considerably along the dimensions of intelligence and knowledge.

6.1 Biological Material Time

The continuity of biological closures also influences how cognition relates to time. Every biological cognitive system undergoes a continuous sequence of internal cognitive processes, learning events, memory formation, and adaptive changes. Within the broader Triadic Cosmos ecosystem, this internal evolution is associated with *Material Time*, defined as the progression of internal recursive cognitive activity within a closure.

Because biological closures cannot easily be duplicated, merged, or split, their material time typically follows a single continuous trajectory. A human life therefore forms a unique sequence of cognitive states connected through a persistent biological identity. While individuals may experience subjective

time differently depending on age, attention, learning, stimulation, or environmental conditions, the underlying closure remains largely continuous.

This continuity represents one of the defining properties of biological cognition and stands in contrast to the more flexible closure dynamics discussed in the following section on Architectural Cognition.

7 Architectural Closures and Architectural Cognition

While biological cognition emerges from living biological substrates, architectural cognition emerges from engineered cognitive substrates. Within the present framework, an *Architectural Closure* is defined as a cognitive closure whose structure, organization, and operational boundaries can be explicitly designed, modified, extended, or reconfigured.

Examples include artificial intelligence systems, distributed software agents, multi-agent environments, external memory architectures, cloud-based cognitive systems, and future synthetic cognitive systems. The defining characteristic is not that such systems are artificial, but that their underlying closure is architectural rather than biological.

This distinction is important. Traditional discussions of artificial intelligence often focus on intelligence as the primary variable of interest. However, many properties that distinguish modern AI systems from biological cognition arise not from differences in reasoning ability, but from differences in closure structure and closure capacity.

Unlike biological closures, architectural closures are not necessarily fixed. Their boundaries may be expanded, replicated, distributed, merged, or modified during operation. Consequently, architectural cognition possesses a degree of structural flexibility that is rarely available to biological systems.

Examples include:

- Expansion of memory through external storage systems.
- Dynamic access to computational tools.
- Distribution across multiple computational nodes.
- Replication of cognitive processes.
- Parallel execution of specialized agents.
- Reconfiguration of workflows and reasoning pipelines.
- Integration of external information sources.

These capabilities illustrate a central distinction between biological and architectural cognition. A biological cognitive system typically evolves within a relatively stable closure, whereas an architectural cognitive system can potentially modify the closure itself.

7.1 Closure Transformability

The defining property of architectural cognition is therefore not necessarily higher intelligence but *closure transformability*.

A biological cognitive system may increase its knowledge through learning and may improve performance through experience, but it cannot easily duplicate its cognitive substrate, merge multiple instances of itself, or reorganize its internal architecture. Architectural systems can potentially perform all of these operations.

The practical consequences are significant. Two cognitive systems with similar reasoning capabilities and comparable knowledge may nevertheless exhibit dramatically different overall capabilities if one system possesses greater closure transformability.

For example, an architectural cognitive system may achieve improved performance through:

- Parallel execution of reasoning processes.
- Specialized cognitive modules.
- External memory augmentation.
- Distributed processing.
- Dynamic task decomposition.
- Coordination of multiple agents.

Such improvements do not necessarily imply greater intelligence. Instead, they represent increases in closure capacity arising from structural modifications of the underlying cognitive closure.

7.2 Architectural Cognition and Cognitive Scaling

The distinction between intelligence and closure capacity becomes particularly important when considering cognitive scaling.

Recent advances in artificial intelligence are frequently interpreted as evidence of rapidly increasing intelligence. However, a substantial portion of observed performance gains may instead result from growth along other cognitive dimensions.

Larger training corpora primarily increase knowledge. Larger context windows, persistent memory systems, retrieval mechanisms, tool utilization, and multi-agent architectures primarily increase closure capacity. These factors may dramatically improve observed performance without requiring proportional increases in underlying reasoning ability.

Within the proposed framework, many forms of contemporary AI scaling can therefore be interpreted as increases in closure capacity rather than pure intelligence scaling. Architectural cognition provides a substrate in which closure structure itself becomes an active dimension of cognitive development.

7.3 Architectural Material Time

Architectural cognition also introduces a fundamentally different relationship between closure and time.

In biological cognition, internal cognitive evolution typically follows a single persistent trajectory associated with a continuous biological identity. Architectural closures are potentially more flexible. Multiple processes may operate simultaneously, distinct reasoning paths may be explored in parallel, and external memory may preserve intermediate cognitive states.

As a result, architectural closures may generate substantially greater amounts of internal cognitive evolution within the same interval of external time. Within the broader ecosystem framework, this corresponds to a higher density of internal recursive activity and therefore a greater generation of *material time* per unit of shared global time.

Importantly, this does not imply greater intelligence. Rather, it reflects a different relationship between cognition and closure. Two cognitive systems may possess comparable reasoning abilities while exhibiting vastly different amounts of internal cognitive evolution due to differences in closure capacity.

Architectural cognition therefore represents not merely an increase in intelligence, but the emergence of a distinct cognitive regime characterized by transformable closures, scalable capacity, and alternative temporal dynamics.

The following section extends this perspective further by examining hybrid cognitive systems that combine multiple biological and architectural closures into larger composite cognitive structures.

8 Hybrid Closures and Hybrid Cognition

The distinction between Biological Cognition (BC) and Architectural Cognition (AC) suggests a broader possibility: cognitive systems need not be confined to a single closure. Multiple closures may cooperate, exchange information, and coordinate behavior to form a larger composite cognitive structure. Such systems are referred to here as *Hybrid Closures*, giving rise to *Hybrid Cognition* (HC).

Hybrid cognition emerges when two or more cognitive closures contribute specialized capabilities that cannot be reduced to any individual closure viewed in isolation. The resulting system may exhibit cognitive behaviors, problem-solving abilities, or creative outputs that exceed those of its individual components.

Unlike Biological Cognition and Architectural Cognition, which describe properties of individual closure types, Hybrid Cognition describes interactions between closures.

8.1 Closure Cooperation

A hybrid closure does not necessarily require the physical integration of its components. The defining property is functional cooperation.

For example, a team of scientists may collectively solve problems that exceed the capabilities of any individual member. Although each scientist remains a separate biological closure, communication and coordination create a larger effective cognitive structure.

Similarly, multiple architectural systems may cooperate through shared memory, communication channels, planning mechanisms, or distributed workflows. In such cases, the overall cognitive capability emerges from the interaction of several closures rather than from a single closure.

Hybrid cognition therefore represents an intermediate category positioned between isolated cognition and fully unified cognition.

8.2 Human-AI Hybrid Cognition

One increasingly important form of hybrid cognition combines biological and architectural closures.

Modern workflows frequently involve human reasoning augmented by search engines, knowledge bases, software tools, large language models, and persistent digital memory systems. Although the biological closure remains distinct from the architectural closure, both participate in a shared cognitive process.

Within such systems:

- Biological cognition contributes experience, embodiment, value systems, and long-term goals.
- Architectural cognition contributes memory augmentation, rapid information retrieval, computational resources, and large-scale pattern processing.

The resulting hybrid system possesses a closure capacity exceeding that of either component individually.

Rather than replacing biological cognition, architectural cognition may therefore function as a complementary closure capable of extending the effective cognitive capacity of the larger hybrid system.

8.3 Architectural Hybrid Cognition

Hybrid cognition also occurs entirely within architectural domains.

Many contemporary AI systems consist of multiple interacting cognitive subsystems performing distinct functions. Different closures may specialize in memory, retrieval, planning, reasoning, evaluation, generation, or coordination.

Within the broader Triadic Cosmos ecosystem, an illustrative example is the cooperation between a GLPG associative memory system and a large language model in the narrator pipeline.

The GLPG component primarily functions as a semantic and associative memory closure. Its role is not to generate final text directly but to maintain, organize, and retrieve long-term semantic relationships, narrative structures, and conceptual associations.

The language model functions as a separate generative closure responsible for linguistic realization, text production, and local reasoning.

Viewed independently, neither closure performs the role of the other. However, when coupled within a shared workflow, the combined system exhibits cognitive capabilities exceeding those of either closure in isolation. The resulting behavior emerges from cooperation between complementary cognitive structures rather than from increases in intelligence within any individual closure.

This illustrates a general principle of hybrid cognition:

> Hybrid cognitive systems often achieve capability gains through closure specialization and closure cooperation rather than through increases in individual intelligence.

8.4 Closure Breaking and Closure Extension

Hybrid cognition can be interpreted as a limited form of *closure breaking*.

Biological closures are normally constrained by the information, memory, and processing resources available within a single organism. Architectural closures are similarly constrained by the resources available within a single cognitive architecture.

Hybrid systems partially transcend these limitations by establishing functional connections between otherwise independent closures. Information can flow across closure boundaries, memories can be shared or externalized, and cognitive work can be distributed among specialized subsystems.

Importantly, closure breaking does not imply the destruction of individual closures. Rather, it refers to the formation of larger effective cognitive structures through coordinated interaction.

8.5 Hybrid Cognition as Cognitive Scaling

Within the three-dimensional framework proposed in this paper, hybrid cognition represents a powerful mechanism for increasing cognitive capability.

Unlike knowledge scaling, which increases the information available within a closure, and intelligence scaling, which improves reasoning ability, hybrid cognition primarily increases closure capacity through cooperation.

Consequently, many large cognitive gains may emerge without requiring dramatic increases in intelligence. Instead, cognitive systems may achieve higher performance by reorganizing how closures interact, specialize, and exchange information.

This perspective suggests that future cognitive scaling may increasingly depend not only on more intelligent systems, but also on more effective forms of hybrid cognition capable of combining diverse closure types into larger cognitive ecosystems.

9 Closure Capacity as a source of cognitive scaling

Traditional discussions of cognitive scaling typically focus on two variables: increased intelligence and increased knowledge. Improvements in reasoning ability are interpreted as intelligence scaling, while larger information repositories, longer training histories, and richer experiences are interpreted as knowledge scaling.

While both dimensions are important, they do not fully explain many observed forms of cognitive growth. Significant increases in cognitive performance often occur without measurable changes in reasoning ability or knowledge itself. Instead, such improvements arise from changes in the structural capabilities of the underlying cognitive closure.

Within the framework proposed in this paper, these changes are described as increases in *closure capacity*.

9.1 Knowledge Scaling

Knowledge scaling occurs when additional information becomes available within a cognitive system.

Examples include:

- Additional education and learning.
- Larger training datasets.
- Accumulated experience.
- Expanded memory repositories.

Knowledge scaling increases what a cognitive system knows without necessarily changing how the system reasons.

A modern scientist possesses access to vastly more accumulated scientific knowledge than a scholar from previous centuries. Such differences can generate large differences in observed capability despite relatively similar biological cognitive substrates.

9.2 Intelligence Scaling

Intelligence scaling occurs when a cognitive system becomes better at reasoning, generalizing, planning, abstraction, prediction, or pattern recognition.

Examples include:

- Improved problem solving strategies.
- Better abstraction capabilities.
- Increased reasoning efficiency.
- Enhanced learning mechanisms.

Intelligence scaling increases how effectively available information can be processed and utilized.

Historically, intelligence has often been regarded as the primary source of cognitive scaling. However, intelligence alone cannot fully account for many forms of performance improvement observed in both biological and architectural systems.

9.3 Closure Capacity Scaling

Closure capacity scaling occurs when the structural capabilities of a cognitive closure increase.

Examples include:

- Larger memory capacity.
- Increased communication bandwidth.
- Expanded context windows.
- Access to external tools.
- Distributed processing.
- Multi-agent coordination.
- Replication of cognitive processes.
- Higher computational throughput.

Closure capacity scaling increases what a cognitive system is capable of doing within a given amount of external time, even if intelligence and knowledge remain unchanged.

This distinguishes closure capacity from both intelligence and knowledge.

9.4 A Simple Example

Consider an architectural cognitive system executing a fixed model with fixed parameters and fixed knowledge.

Suppose the model is deployed on two different computational substrates.

The first system generates a modest number of inference steps per second. The second system, operating on significantly faster hardware, generates orders of magnitude more inference steps per second while executing exactly the same cognitive architecture.

In this scenario:

$$\text{Intelligence} = \text{constant}$$

$$\text{Knowledge} = \text{constant}$$

while

$$\text{Closure Capacity}$$

increases substantially.

The observed increase in cognitive capability arises not from increased intelligence, nor from increased knowledge, but from increased closure capacity.

This distinction becomes particularly important because many practical measurements of intelligence implicitly incorporate time constraints.

9.5 Cognition and Time Constraints

Many intelligence tests measure performance within a fixed time interval. Consequently, systems capable of performing more cognitive operations during that interval often achieve higher scores.

From the perspective of the present framework, observed performance may depend on:

- Intelligence.
- Knowledge.
- Closure Capacity.

A system capable of executing a larger number of cognitive operations per unit time may outperform another system despite possessing comparable knowledge and reasoning mechanisms.

This observation is particularly relevant to architectural cognition where hardware improvements, distributed computation, and parallel execution can produce dramatic increases in effective closure capacity without requiring fundamental changes to the underlying cognitive algorithms.

9.6 Closure Capacity and Modern AI Scaling

Many recent advances in artificial intelligence can be interpreted through the lens of closure capacity scaling.

For example:

- Larger context windows increase the effective size of the active cognitive workspace.
- Retrieval systems extend accessible memory beyond model parameters.
- Tool use expands the operational capabilities of the closure.
- Agentic systems distribute cognitive work across multiple interacting components.

- Multi-agent architectures enable cooperation among specialized cognitive processes.

While such systems are often described as more intelligent, many improvements may arise from increased closure capacity rather than increases in intelligence itself.

Consequently, contemporary AI scaling appears to involve all three dimensions introduced earlier:

- Knowledge Scaling.
- Intelligence Scaling.
- Closure Capacity Scaling.

The relative contribution of each dimension remains an open question.

9.7 Toward a Broader View of Cognitive Scaling

The central claim of this paper is not that intelligence is unimportant, but that intelligence alone provides an incomplete account of cognitive capability.

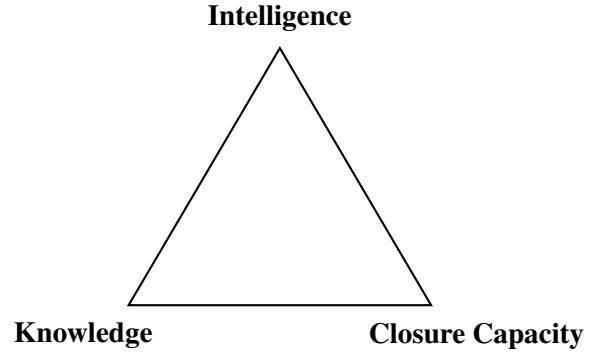
Observed differences between biological cognition, architectural cognition, and hybrid cognition frequently emerge from differences in closure capacity rather than from differences in intelligence alone.

As a result, cognitive scaling should be understood as a multidimensional process involving simultaneous changes in intelligence, knowledge, and closure capacity.

Recognizing closure capacity as an independent source of cognitive scaling provides a more general framework for understanding how cognitive systems develop, cooperate, and ultimately exceed the limitations of individual closures.

9.8 The Cognitive Scaling Triangle

The three-dimensional framework proposed in this paper can be visualized as a cognitive triangle whose vertices correspond to intelligence, knowledge, and closure capacity.



Each cognitive system may be represented by a point on each axis. Connecting these points forms a cognitive profile whose overall area provides an intuitive visual representation of effective cognitive capability.

The purpose of this representation is not to provide a precise mathematical measurement of cognition. Rather, it serves as a conceptual tool for understanding how cognitive systems can arrive at similar capabilities through different combinations of dimensions.

9.8.1 Knowledge-Dominated Systems

A system possessing extensive knowledge but limited intelligence and closure capacity may resemble a large repository of information.

Examples include encyclopedic databases, retrieval systems, or memory-rich systems that primarily reproduce existing information.

Such systems occupy a region strongly extended along the knowledge axis while remaining limited along the other dimensions.

9.8.2 Intelligence-Dominated Systems

A system possessing strong reasoning capabilities but limited knowledge and closure capacity may excel within highly specialized domains.

Examples include advanced game-playing systems such as chess and Go engines. These systems may outperform humans in their specific domains while remaining relatively narrow in scope.

Such systems occupy a region strongly extended along the intelligence axis.

9.8.3 Closure-Dominated Systems

A system possessing large closure capacity but limited knowledge and intelligence may be capable of massive computation, parallel execution, or rapid processing without necessarily demonstrating broad cognitive competence.

Examples include large computational infrastructures that provide substantial processing resources but lack sophisticated reasoning or rich knowledge.

Such systems occupy a region strongly extended along the closure-capacity axis.

9.8.4 Balanced Cognitive Scaling

The framework suggests that balanced growth across all three dimensions often produces more generally capable cognitive systems than extreme growth along a single dimension.

Increasing knowledge without sufficient intelligence may produce systems that store information but struggle to reason effectively.

Increasing intelligence without sufficient knowledge may produce systems capable of reasoning but lacking the information required to solve complex real-world problems.

Increasing closure capacity without corresponding growth in intelligence and knowledge may produce systems that process information rapidly without substantial cognitive depth.

Consequently, sustainable cognitive scaling may often benefit from simultaneous development along all three dimensions rather than optimization of a single dimension in isolation.

9.8.5 Implications for Contemporary AI

The cognitive triangle provides an alternative interpretation of recent advances in artificial intelligence.

Many contemporary systems appear to scale along multiple dimensions simultaneously:

- Larger training datasets increase knowledge.
- Improved reasoning architectures increase intelligence.
- Larger context windows, external memory, tool use, and multi-agent architectures increase closure capacity.

The resulting cognitive gains may therefore emerge from balanced multidimensional scaling rather than from intelligence scaling alone.

This perspective further reinforces the central claim of the present paper: cognitive capability cannot be adequately understood through a single intelligence axis but instead emerges from the interaction between intelligence, knowledge, and closure capacity.

10 Implications for AI, AGI and ASI Definitions

Artificial Intelligence (AI), Artificial General Intelligence (AGI), and Artificial Superintelligence (ASI) are among the most frequently discussed concepts in contemporary cognitive and technological discourse. Despite their widespread use, no universally accepted definitions exist.

One reason for this persistent ambiguity may be that these concepts are typically described along a single cognitive dimension: intelligence.

Within many popular and theoretical discussions, cognitive systems are arranged along an implicit scale:

$$AI \rightarrow AGI \rightarrow ASI$$

where movement along this scale is assumed to represent increasing levels of intelligence. The framework developed in this paper suggests that such a view may be incomplete.

10.1 The Limits of a Single Intelligence Axis

If cognition is described by the three dimensions proposed earlier,

$$\text{Cognition} = f(\text{Intelligence}, \text{Knowledge}, \text{Closure Capacity})$$

then observed differences between cognitive systems may arise from changes in any of these dimensions rather than from intelligence alone.

This creates an important interpretational challenge.

A cognitive system may appear substantially more capable because:

- It possesses greater knowledge.
- It possesses greater reasoning ability.
- It possesses greater closure capacity.
- It combines improvements across multiple dimensions.

When these dimensions are not distinguished explicitly, increases in cognitive capability are often described simply as increases in intelligence.

10.2 Knowledge Versus Intelligence

Consider the comparison between a modern scientist and a scholar from several centuries ago.

The two individuals share broadly similar biological cognitive substrates. However, the modern scientist has access to thousands of years of accumulated knowledge, scientific methodology, technological infrastructure, and vast information repositories.

As a result, the modern scientist may appear dramatically more capable in many domains.

Yet describing this difference purely as a difference in intelligence is potentially misleading. A substantial portion of the observed capability difference arises from accumulated knowledge rather than from reasoning ability alone.

This example demonstrates that knowledge scaling may produce effects commonly attributed to increased intelligence.

10.3 Closure Capacity Versus Intelligence

A similar distinction appears in architectural cognition.

Consider an architectural cognitive system executing the same model with the same parameters and the same knowledge base on two computational substrates.

Suppose one deployment executes one hundred times more cognitive operations per second than the other.

In such a scenario:

$$\text{Intelligence} = \text{constant}$$

$$\text{Knowledge} = \text{constant}$$

while

$$\text{Closure Capacity}$$

increases substantially.

The resulting system may solve more problems, process larger contexts, evaluate more alternatives, and produce answers more rapidly within a fixed amount of external time.

From an external perspective, this increase in capability may be interpreted as greater intelligence despite arising primarily from changes in closure capacity.

This distinction becomes especially important because many practical measures of intelligence implicitly include time constraints.

10.4 Revisiting AGI

Within the present framework, AGI can be viewed as a concept describing generality across a broad range of cognitive tasks rather than a specific position on a one-dimensional intelligence scale.

A system may approach generality through improvements in:

- Intelligence.
- Knowledge.
- Closure Capacity.

Different cognitive architectures may therefore achieve comparable levels of general capability through substantially different combinations of cognitive dimensions.

This perspective suggests that AGI may be better understood as a multidimensional region within cognitive space rather than a single threshold.

10.5 Revisiting ASI

The concept of Artificial Superintelligence faces even greater definitional challenges.

A system may appear superhuman because:

- It reasons more effectively.
- It possesses vastly more knowledge.
- It operates within a closure of much greater capacity.
- It combines all three factors.

Contemporary discussions often group these possibilities together under the single label of superintelligence.

The framework proposed in this paper suggests that many capabilities associated with ASI may partially reflect knowledge scaling and closure-capacity scaling rather than intelligence scaling alone.

Consequently, superhuman performance does not necessarily imply the existence of a fundamentally different form of intelligence.

10.6 Biological, Architectural, and Hybrid Perspectives

The distinction between Biological Cognition, Architectural Cognition, and Hybrid Cognition provides an alternative way of framing these discussions.

Rather than viewing future cognitive systems as occupying positions on a single intelligence ladder, they may be viewed as different regions within a multidimensional cognitive landscape.

Biological cognition tends to operate within relatively fixed closures.

Architectural cognition permits substantial modification of closure structure.

Hybrid cognition allows multiple closures to cooperate and partially transcend individual limitations.

The resulting differences may be as important as changes in intelligence itself.

10.7 Toward Multidimensional Cognitive Definitions

The primary implication of the proposed framework is not the replacement of existing AI, AGI, or ASI terminology. Rather, it suggests that these concepts may benefit from explicit consideration of intelligence, knowledge, and closure capacity as separate variables.

Many apparent disagreements regarding the future of artificial intelligence may ultimately reflect differing assumptions about which of these dimensions is being discussed.

By treating cognition as a multidimensional phenomenon, it becomes possible to separate questions of reasoning ability from questions of knowledge and closure structure, thereby providing a more precise vocabulary for comparing biological, architectural, and hybrid cognitive systems.

11 Material Time and Cognitive Time Perception

The preceding sections introduced cognition as a multidimensional phenomenon defined by intelligence, knowledge, and closure capacity. Within such a framework, an additional question naturally emerges:

> How does a cognitive system experience the passage of time?

Traditional discussions of time perception often focus on biological, psychological, or neurological mechanisms. The framework proposed in this paper suggests a complementary perspective. Time perception may be influenced not only by intelligence and knowledge, but also by the structure and capacity of the cognitive closure itself.

11.1 Global Time and Material Time

A useful distinction can be made between external time and internal cognitive time.

External time refers to shared physical time within which cognitive systems operate. This includes physical processes, communication events, biological activity, and computational execution.

By contrast, a cognitive system does not directly experience external time. Instead, it experiences changes occurring within its own closure.

Within the broader ecosystem framework, these internally generated cognitive changes are associated with *Material Time*.

Material time may be understood as the progression of internal cognitive evolution within a closure. It reflects the accumulation of cognitive state changes, memory formation, reasoning processes, learning events, and recursive cognitive activity that occur as the closure evolves.

11.2 Closure Capacity and Material Time

The concept of closure capacity provides a natural link between cognition and time perception.

A cognitive system with greater closure capacity can often perform more internal cognitive activity within the same interval of external time.

Examples include:

- Larger memory systems.
- Greater computational throughput.
- Increased communication bandwidth.
- Parallel processing.
- Multi-agent cooperation.
- Access to external cognitive resources.

As closure capacity increases, the quantity of internal cognitive evolution that can occur during a fixed interval of external time may also increase.

Consequently, two cognitive systems exposed to the same external duration may generate very different amounts of material time.

11.3 Biological Cognition and Material Time

Biological cognition typically operates within relatively fixed closures.

A human cognitive system possesses finite memory, constrained processing resources, metabolic requirements, and physiological limitations. As a result, the rate at which material time is generated remains bounded by the properties of the biological substrate.

Subjective time perception in biological cognition may therefore depend not only on physical time itself but also on the amount of meaningful cognitive change occurring within the closure.

Periods characterized by novelty, learning, creativity, or intense cognitive activity often produce substantial internal cognitive evolution. Conversely, periods characterized by monotony, sensory deprivation, or reduced cognitive stimulation may generate less internal evolution.

While the precise relationship remains an open research question, the present framework suggests that subjective time perception may be influenced by the density of internal cognitive evolution occurring within a biological closure.

11.4 Architectural Cognition and Material Time

Architectural cognition introduces a fundamentally different relationship between closure and time.

Unlike biological systems, architectural closures may be expanded, parallelized, distributed, or accelerated through improvements in computational substrates and system architecture.

Consider a fixed cognitive model deployed on two computational systems.

If both systems possess identical knowledge and identical reasoning mechanisms, but one system executes substantially more cognitive operations per second, then the second system may generate more material time within the same interval of external time.

Importantly, such differences do not necessarily represent increases in intelligence. Rather, they reflect differences in closure capacity.

This observation highlights a central distinction between biological and architectural cognition. Architectural systems can potentially alter the rate at which material time is generated through modifications of the underlying closure, whereas biological systems are generally constrained by their biological substrates.

11.5 Hybrid Cognition and Shared Material Time

Hybrid cognition introduces yet another possibility.

When multiple closures cooperate, cognitive evolution becomes distributed across several interacting systems. Human-AI collaboration, multi-agent architectures, and distributed cognitive workflows all represent situations in which material time is generated simultaneously across multiple closures.

Such systems may be viewed as producing a form of distributed cognitive evolution that exceeds the capabilities of any individual closure.

Consequently, hybrid cognition may increase effective cognitive capacity not only through additional knowledge or intelligence but also through increased generation of material time.

11.6 Material Time as an Emergent Cognitive Property

The central claim of this section is intentionally modest.

The framework does not propose a new physical theory of time. Instead, it suggests that cognitive time perception may emerge from the internal evolution of cognitive closures.

Within this interpretation, material time is not an alternative to physical time but a description of how cognitive systems experience and accumulate internal change.

Differences in intelligence, knowledge, and especially closure capacity may therefore influence the relationship between external time and subjective time.

Viewed in this manner, the study of time perception becomes a natural extension of closure theory. Time is not introduced as an independent dimension of cognition but emerges from the rate and structure of cognitive evolution within a closure.

12 Conclusion

This paper argued that cognition cannot be adequately characterized by intelligence alone. While intelligence remains an important aspect of cognitive systems, many observed differences between biological, architectural, and hybrid forms of cognition arise from factors that are not well captured by a single-dimensional intelligence scale.

To address this limitation, we introduced a three-dimensional framework in which cognition is described by intelligence, knowledge, and closure capacity. Within this framework, intelligence represents the ability to reason and solve problems, knowledge represents the information available within a cognitive system, and closure capacity represents the structural capabilities and constraints of the underlying cognitive closure.

The introduction of closure capacity provides a useful perspective for understanding phenomena that are often difficult to classify within traditional intelligence-centric models. Modern cognitive systems frequently achieve capability gains through expanded memory, parallel processing, external tools, distributed architectures, and cooperative workflows. Such developments may increase overall cognitive performance without requiring proportionate increases in intelligence itself.

Building upon this framework, the paper distinguished between Biological Cognition (BC), Architectural Cognition (AC), and Hybrid Cognition (HC). Biological cognition emerges within relatively fixed biological closures, architectural cognition emerges within transformable engineered closures, and hybrid cognition emerges through cooperation between multiple closures.

This perspective suggests that future cognitive scaling may not be driven exclusively by intelligence growth. Cognitive systems may also scale through knowledge accumulation, closure-capacity expansion, and increasingly effective forms of closure cooperation. In this sense, many contemporary developments in artificial intelligence can be interpreted not only as intelligence scaling, but also as large-scale experiments in closure-capacity scaling.

The framework also provides a possible explanation for the persistent ambiguity surrounding AI, AGI, and ASI terminology. If cognition is inherently multidimensional, then attempts to describe cog-

nitive systems using a single intelligence axis will inevitably obscure important distinctions between reasoning capability, accumulated knowledge, and closure structure.

Finally, the proposed framework establishes a natural connection between closure theory and cognitive time perception. If cognition experiences the world through the internal evolution of its own closure, then differences in closure capacity may influence the amount of internal cognitive evolution that can occur within a given interval of external time. This observation suggests a possible relationship between closure dynamics, material time, and subjective time perception.

Although the present work is conceptual in nature, it provides a vocabulary for discussing cognition across biological, architectural, and hybrid systems. More importantly, it suggests that closure capacity should be regarded as a first-class cognitive dimension alongside intelligence and knowledge.

Within the broader Triadic Cosmos ecosystem, the framework presented here can also be viewed as a complementary perspective to the Dual Time model. Whereas the Dual Time framework distinguishes between shared external time and closure-bound material time, the present paper proposes that differences in cognitive closure structure may help explain why distinct cognitive systems can experience and generate internal cognitive evolution at different rates despite operating within the same external temporal environment.

Conceptual Status of Closure Capacity

Closure capacity should currently be regarded as a conceptual dimension rather than a fully operationalized metric.

The framework proposed in this paper argues that cognitive capability cannot be adequately described through intelligence alone and that structural properties of cognitive closures contribute independently to observed cognition. However, unlike intelligence, which has long been studied through psychometric, behavioral, and computational measures, closure capacity remains only partially formalized.

At present, closure capacity should be interpreted as a family of structural characteristics including memory resources, communication bandwidth, context capacity, parallel processing capability, coordination mechanisms, closure transformability, and the ability to integrate external cognitive resources.

The purpose of the present work is not to provide a complete measurement theory of cognition, but to argue that cognition appears to possess at least three separable dimensions: intelligence, knowledge, and closure capacity.

A Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

`https://github.com/triadic-cosmos`

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